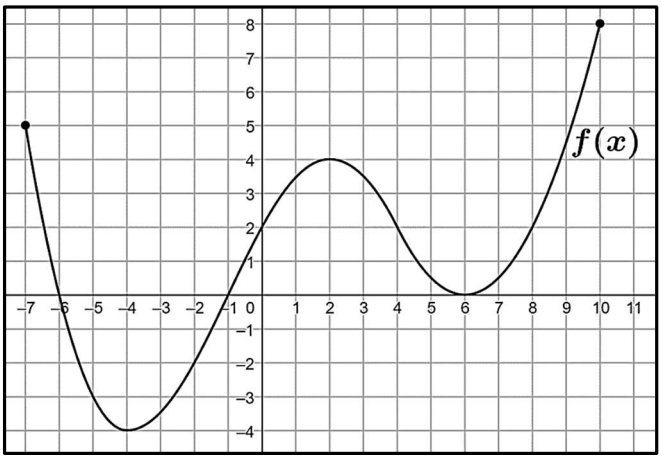
Worksheet 5-9 A Connecting f, f', and f'' #1
AP Calculus Name_____

Directions: The graph of $f(x)$ is given below. Use the graph to answer the following questions. If not enough information is given to answer a question, write "Cannot be determined".



1. At which x value(s) is $f(x) = 0$?

2. At which x value(s) is $f'(x) = 0$?

3. On what interval(s) is $f(x)$ increasing?

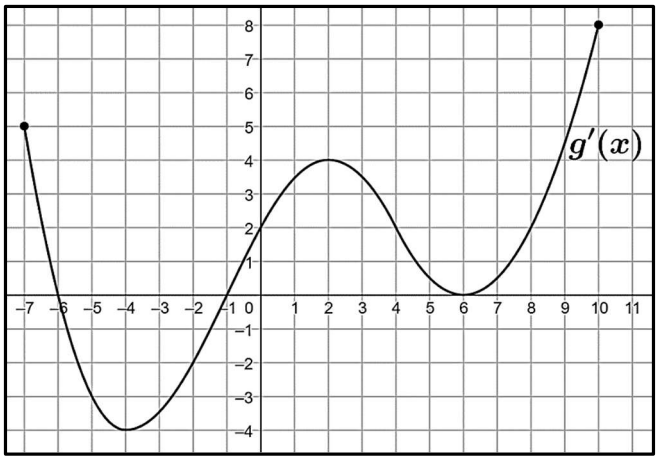
4. At which x value(s) does $f(x)$ have a relative minimum? .

5. At which x value(s) does $f''(x) = 0$?

6. On what interval(s) is $f(x)$ concave down?

7. At which x value(s) does $f(x)$ have a point of inflection? Explain your reasoning.

Directions: The graph of $g'(x)$ is given below. Use the graph to answer the following questions. If not enough information is given to answer a question, write "Cannot be determined".



8. At which x value(s) is $g(x) = 0$? Explain your reasoning.

9. At which x value(s) is $g'(x) = 0$?

10. On what interval(s) is $g(x)$ increasing? Explain your reasoning.

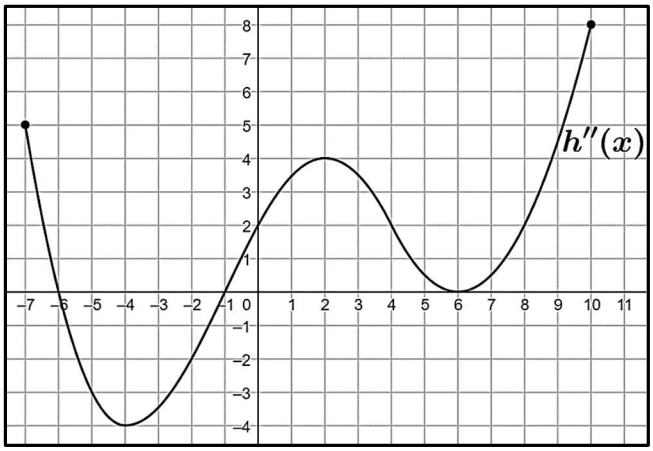
11. At which x value(s) does $g(x)$ have a relative minimum? Explain your reasoning.

12. At which x value(s) does $g''(x) = 0$? Explain your reasoning.

13. On what interval(s) is $g(x)$ concave down? Explain your reasoning.

14. At which x value(s) does $g(x)$ have a point of inflection? Explain your reasoning.

Directions: The graph of $h''(x)$ is given below. Use the graph to answer the following questions. If not enough information is given to answer a question, write "Cannot be determined".



15. At which x value(s) is $h(x) = 0$? Explain your reasoning.

16. At which x value(s) is $h'(x) = 0$? Explain your reasoning.

17. On what interval(s) is $h(x)$ increasing? Explain your reasoning.

18. At which x value(s) does $h(x)$ have a relative minimum? Explain your reasoning.

19. At which x value(s) does $h''(x) = 0$? Explain your reasoning.

20. On what interval(s) is $h(x)$ concave down? Explain your reasoning.

21. At which x value(s) does $h(x)$ have a point of inflection? Explain your reasoning.

For each table, selected values of x and $f(x)$ are given. Assume that $f'(x)$ and $f''(x)$ do not change signs. Answer the questions for each table.

22.)

x	$f(x)$
4	-5
5	-8
6	-12
7	-17

a.) is $f(x)$ increasing or decreasing?

b.) is $f(x)$ concave up or concave down?

23.)

x	$f(x)$
-3	-2
-2	3
-1	7
0	10

a.) is $f(x)$ increasing or decreasing?

b.) is $f(x)$ concave up or concave down?

24.)

x	$f(x)$
2	3
3	0
4	-2
5	-3

a.) is $f(x)$ increasing or decreasing?

b.) is $f(x)$ concave up or concave down?

25.)

x	$f(x)$
0	1
1	2
2	4
3	7

a.) is $f(x)$ increasing or decreasing?

b.) is $f(x)$ concave up or concave down?